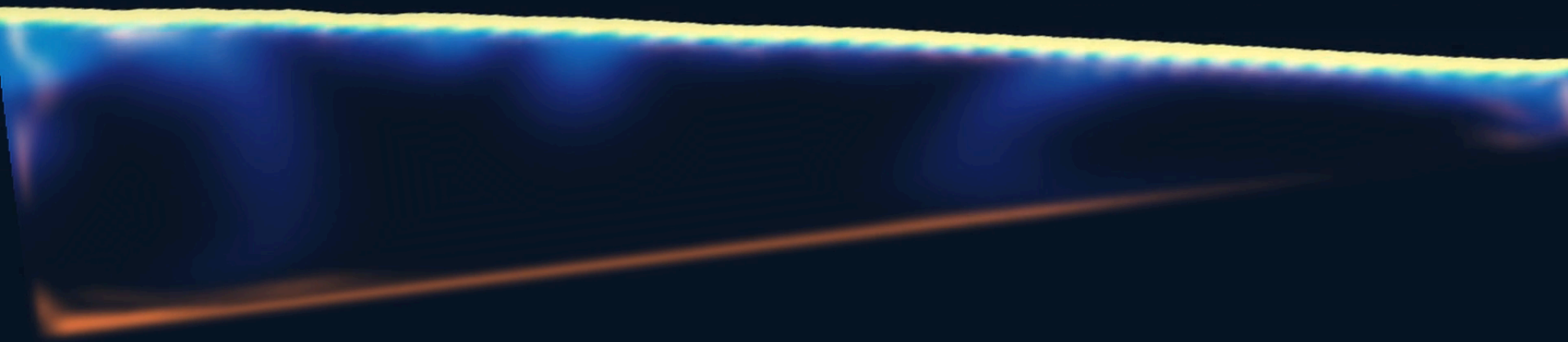
Multi-fidelity Bayesian optimization of mixing in two-phase rocking bioreactors

Elvis Aguero · [co-authors] · Brown University

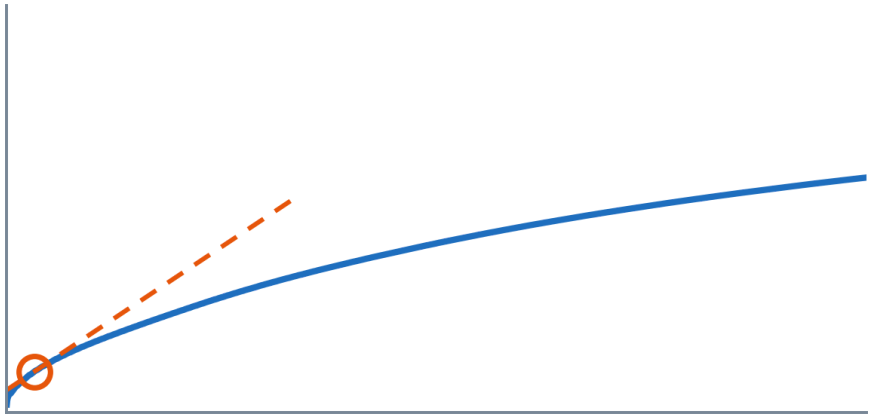


Motivation

Introduction

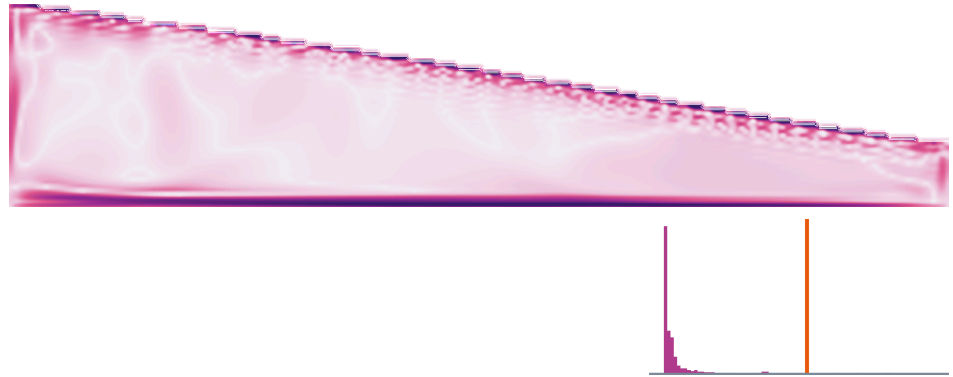
kLa

$$dC^*/dt = kLa\,(1 - C^*)$$



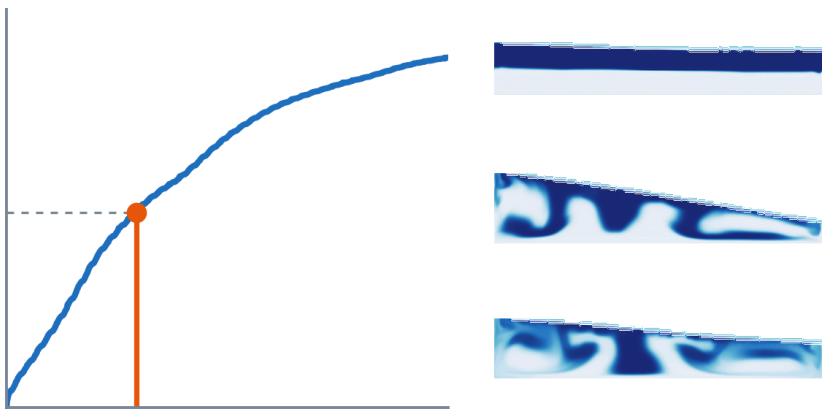
shear stress

$$\tau = \mu\,(\partial u/\partial y + \partial v/\partial x)$$



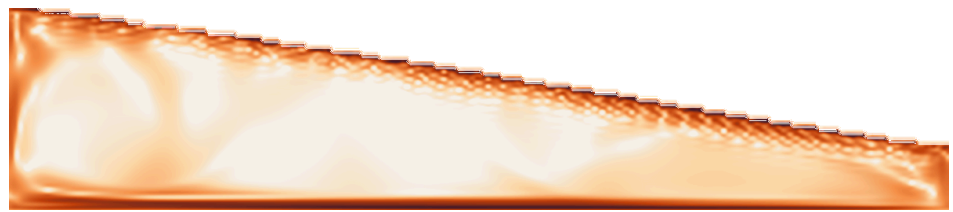
mixing time

$$\chi = 1 - \sigma^2 / \sigma^2_{max}$$



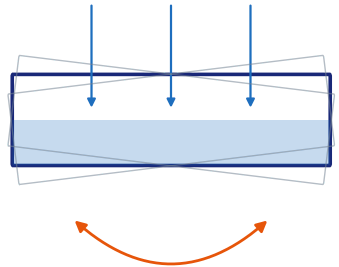
dissipation rate

$$\varepsilon = 2\,\mu\,S : S$$



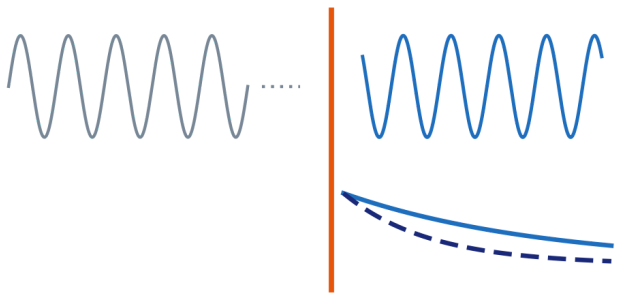
Basilisk

Navier–Stokes + VOF

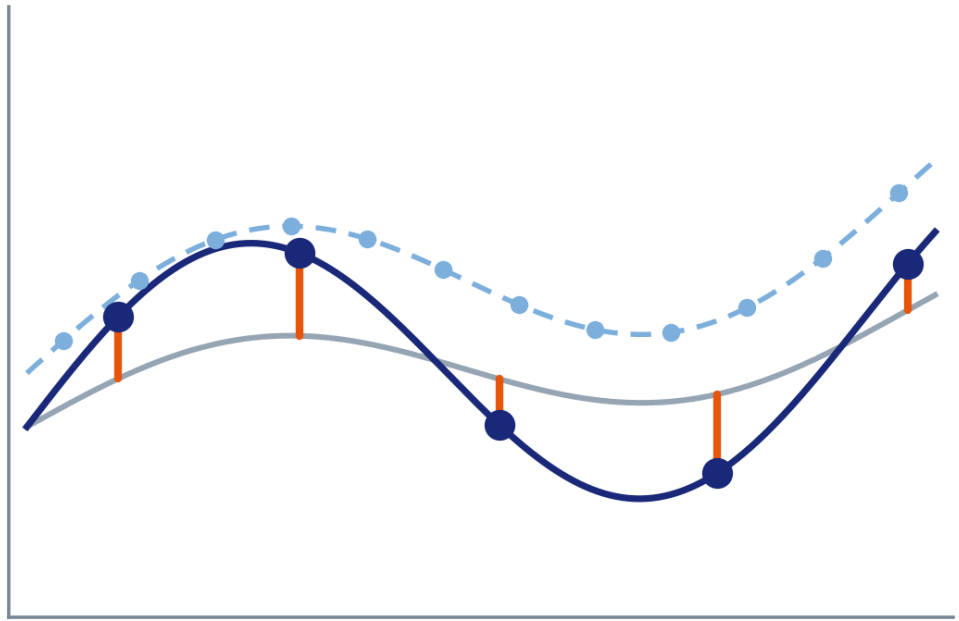


release

$$\partial c/\partial t + u\cdot\nabla c = \nabla\cdot(D\nabla c)$$



multi-fidelity



$$f_{\mathrm{H}}(x) = \rho\, f_{\mathrm{L}}(x) + \delta(x)$$

$\delta \sim \mathrm{GP}$

Results 1

Results 2

Conclusions

Questions?